agentAI: A Research-Data Platform for Physical and Embodied Agents

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A researcher states a goal; a planning model proposes the project and its agents, and the researcher corrects it. They then run in a sandbox, on data they harvest, take from a robot as it drives, or receive by upload, and train on it round after round unwatched. A loop like that is only as good as what it reports about itself. This sheet is that coordination, and the review that makes it checkable.

Weed Detection



The corpus exists to train the weed detector the laser robot needs. Two agents work it unprompted: one harvests and audits public imagery, the other trains. Over 150,000 labelled images.

Rover



A four-wheel skid-steer narrow enough for a crop row, carrying its own battery and compute. Every drive it opens becomes a governed dataset as it runs: camera, GPS and two inertial units, over one API key.

Laser Weed Control Robot



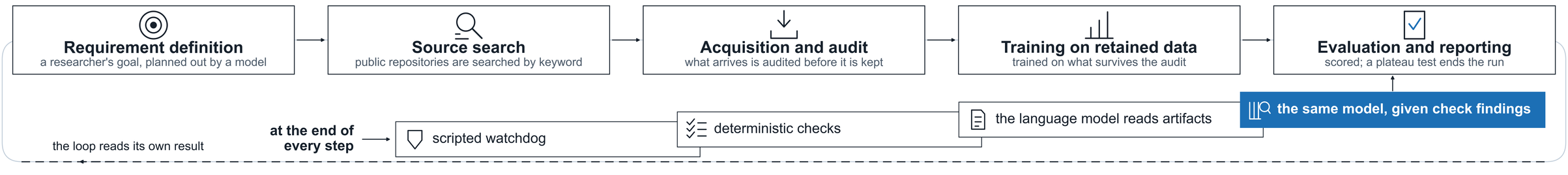
The end use the collection is for. A YOLO detector localises each weed and two galvanometer lasers aim at it and fire, destroying it in place—no chemical, and nothing touches the crop beside it.

Humanoid and Quadruped



Legs reach ground wheels cannot, and in the field this one crossed it faster than the cart. A wide stance and a level gait hold the camera steady, which is what a collector needs.

Four projects, one platform. Each was opened by the same procedure, and its data is registered the same way whether it was harvested, driven in by a robot, or uploaded.



Agent Workflow Diagram

Figure 1. The harvest-and-train loop, as it runs unattended. A researcher's goal is planned into keyword queries, what they return is audited, what survives is trained on, and the round is scored. The stopping rule is a plateau test today and a reviewer's judgement by design. Review follows every step.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

agentAI is one platform for every laboratory robot

A robot, drone or sensor rig is a project: its own dataset registry, its own agents, and an owner who controls both. Agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.

The weed detector runs on a robot's live view

While a robot drives, its project page shows the newest camera frame every second and a half. One switch runs the campaign's own weed detector over that frame, so the operator sees what the model sees, on the drive that is happening.

Ask any dataset a question in plain words

Every dataset page has a chat box. A planner reads the question and the data's real columns, then picks a built-in analysis or writes new code. That code runs in a sandbox with no network and no file writes, and the answer returns with its chart and code.

Every data type and domain forms its own project

On upload, a project returns an automatic diagnosis of its data, images or sensor logs alike, before anyone chooses a next step. The diagnosis is where the planner starts when the researcher asks what the data can support.

The platform is live; the QR code links to it

lab-b660m-c.tailfa6424.ts.net
Sign in with a Google account to open it.

OPEN-WEIGHT MODELS REVIEW IN TIERS

Citing evidence beats model size

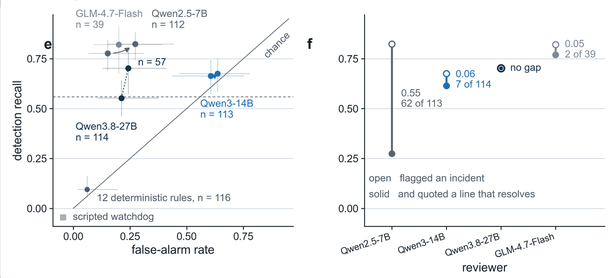


Figure 2. (e) Recall and (f) grounded recall on 149 frozen failures taken from this project's own record, scored by the project's own scorer. Each arrow runs from the model reading the raw files to the same model handed the twelve checks' findings. Grounded counts only the cases where the model also quoted a line that resolves the failure. The dotted arrow carries no head: its two ends were scored on different cases, 57 against 114.

We froze 149 real failures from our own history and asked open-weight models to find them. Good reviewers cite their evidence; size matters less. Handed the twelve checks' findings, a 27-billion model found 0.149 more, a 14-billion one 0.012.

Review runs even when nothing looks wrong

Review follows every step: failures here announced themselves as successes. Four levels by cost: the running watchdog, twelve fixed checks, a model reading files, then that model given the findings. The watchdog returned nothing on 149 frozen failures.

WHEN A MODEL'S JUDGEMENT COUNTS

What the lab model says stays a draft

On the cluster	
plan the next step	qwen3.8 27B
review in depth	glm-4.7-flash
hard one-off reasoning	deepseek-v3 671B
decide what to collect	gemma 4
judge dataset quality	qwen2.5 7B
label assistance	minicpm-v
On the lab box	
turn intent into a project	qwen2.5 3B
summarise a dataset	qwen2.5 3B
The tiers above them	
the hardest open problems	Kimi K3, largest open
check and correct a campaign	Claude Opus 5, via Claude Code
plan the next experiments	Claude Fable 5.1, own key

Figure 3. The eight jobs that need a model and the model each was given, read from the router's own table; a filled square marks a job that exercises judgement. The dashed band is the escalation above them.

Eight jobs need a model. Most run on the cluster; two run in the lab on a 3B model, because a person is waiting. A lab answer is marked so it cannot count as a decision: the rule about where a judgement counts travels with the answer, not with the configuration.

A model's idea was tested before it was used

Downloads carry no plant name, so every box from a site shares one invented label. A model named that as the cause and wrote down first what would disprove it; a human approved, the first decision to pass the full approval chain. Four runs disproved it.

WHEN A MODEL'S JUDGEMENT COUNTS (CONT.)

Why a loop's own report cannot count

Two programs ran unattended on the cluster, one collecting and one training, for fifteen rounds. After the third the training set never changed, yet each round was logged as new work: a job that finished was recorded as progress, whatever it produced.

WHAT THE COLLECTION PRODUCED

156,521 labelled by machine, 3,208 kept

A machine labelled 156,521 pictures; 59,134 remain once duplicate downloads are removed. We audited 13,527 of them, and only 3,208 came from a source worth keeping. The checker scores 1.000 on hand-labelled pictures, so a low score means the source is poor.

Gemma 4 said yes to every box

We cropped each box out and asked Gemma 4, a vision model, whether it held a weed. It said yes every time, wrong ones included, so it cannot serve as a check. Distance from trusted examples, on DINOv2 features, does the work. Fifty-one datasets were cut.

One bad box hides from a dataset filter

A labelling mistake is usually one bad box in an otherwise sound picture, invisible to a whole-dataset filter. The check has to run box by box. Those tools are built, and what gating the merge on their scores is worth is the next experiment.